



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT (Part II)

Date: 01/08/2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No. : PHA G538
Course Title : Immunopharmacology
Instructor-in-Charge : Aniruddha Roy

1. Scope and Objective of the Course:

This course covers key immunologic concepts: how the components of the immune system are generated, organized, and function; the mechanisms that lead to beneficial immune responses, immune disorders, and immunodeficiencies; methods to identify immune reactions; and ways to manipulate the immune system and its components to benefit the patient. This course will give the students an advanced understanding of the principles and mechanisms of the immune system and immune responses in the context of infection, malignancy and immunological disorders. The course will also enable the students to describe the fundamental mechanisms underlying immunologic disease and associate these mechanisms with strategies for therapeutic modulation of the immune system. It will help the students to gain the necessary transferable and research skills in basic and clinical immunology to promote lifelong learning and career development.

2. Text Book :

1. Kuby Immunology by Owen et al., 7th Ed. Freeman press. 2013.

3. Reference Books :

1. Janeway's Immunobiology, Eighth Edition.
2. Cellular and Molecular Immunology, Abul K. Abbas, Eighth Edition.

4. Course Plan

Lec. No.	Topics	Contents	Ref.
1-3	Introduction to Immunology	Cells of the immune system, innate and acquired immunity, primary and secondary lymphoid organs, antigens: chemical and molecular nature, clonal selection theory, humoral and cellular immunity.	TB 1: Ch 1, Ch 2, Ch 3, Ch 4, Ch 5
4-6	Humoral Immunity	B-lymphocytes and their activation, structure and function of immunoglobulins, antibody genes and generation of diversity, production of monoclonal antibodies and applications, cytokines.	TB 1: Ch 5, Ch 11, Ch 12
7-10	Cell-mediated Immunity	Activation and function of T-cells, antigen presenting cells, antigen processing and presentation, Major histocompatibility Complex- MHC Class I and II molecules.	TB 1: Ch 9, Ch 7, Ch 8
11-20	Infection and Immunity	Injury and inflammation, immune response to infections: immunity to viruses, bacteria, fungi and parasites, complement, immunosuppression, tolerance, allergy and hypersensitivity, AIDS and immunodeficiencies, resistance and immunization.	TB 1: Ch 17, Ch 16, Ch 19



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21-25	Vaccinology	Development of Live attenuated, Killed, sub-unit, recombinant vaccines and their use.	TB 1: Ch 18
26-28	Transplantation immunology	Relationship between donor and recipient, role of MHC molecules in allograft rejection, bone marrow and haematopoietic stem cell transplantation.	TB 1: Ch 21
29-32	Tumor immunology	Tumor antigens, categories of tumor antigens, tumor immuno editing hypothesis, tumor mediated immune suppression, immune therapy of tumor.	TB 1: Ch 22
33-35	Autoimmunity	Criteria and causes of autoimmune disorders, autoimmune haemolytic anemia, myasthenia gravis, systemic lupus erythematosus, multiple sclerosis, rheumatoid arthritis.	TB 1: Ch 20
36-38	Immunodeficiency diseases, allergy and hypersensitivity reactions	Definition and origin. Immunological characteristics of primary and secondary immune deficiencies. Syndromes and diseases. Allergy. Terms. Types of allergens. Types of allergic reactions –characteristics, mechanism of tissue damage, diseases and syndromes –examples.	TB 1: Ch 16
39-40	Immunodiagnostics	Evaluating immunological functions, Double diffusion, single radial immunodiffusion, ELISA, radioimmunoassay, immunofluorescence, fluorescence activated cell sorting analysis, HLA typing and matching, cytotoxicity assay.	TB 1: Ch 23, Class notes.

5. Evaluation:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-Semester Test	90 Min.	30	<TEST_1>	CB
Comprehensive Examination	3 h	40	<TEST_C>	CB + OB
*Continuous Assessment		30	During semester	

*Continuous assessment will be based on theory covered in class. Topics and number will be announced in the class. It will be in terms of home assignments, tutorials, and surprise tests.

6. Grading Procedure: Grading would be done by the bunching procedure. In borderline cases subjective judgment will be used to award the grades. It is not mandatory to award all the eight grades (i.e. from A to E). Subjective judgment based on attendance for the lectures, tutorials, appearance in quiz, student's involvement in the course and performance in the class would be used in the award of grades. **The student shall not be considered as "exposed" to the course, unless he/she demonstrates appreciable skill in both the class and theory components of the course and through classroom participation.** Attending tutorial and appearing surprise quiz is very important. Students not appeared in any quiz or not submitted assignments, not appeared in any test/ comprehensive exam will not be considered exposed to the course. It is also expected that student will attend classes regularly to get proper exposure and to provide chance of evaluation of his knowledge.